BASIL AYYAD

Software Engineer · Backend

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PROFESSIONAL SUMMARY

Software Engineer focused on backend systems and AI agents, with hands on experience building REST APIs, async services, and full stack products across web, mobile, and desktop. Worked on PolarisX ERP at Obsidian Labs, a startup in its early stages. Also built Arjan, an AI assisted healthcare platform, as a graduation project. Closely follows the global ERP market through hands on demos of leading products.

PROFESSIONAL EXPERIENCE

Software Engineer · Obsidian Labs (Early Stage Startup) *Dec 2025 to Oct 2026*

- Played a key role in the ongoing development and improvement of PolarisX ERP, enhancing existing features and implementing new functionality across inventory, CRM, POS, accounting, reporting, and document management modules.
- Contributed to backend API development with Sanic, system integrations, and cross platform functionality.
- Improved performance, efficiency, and reliability through code optimization, debugging, and continuous refinement of systems.
- Identified, investigated, and resolved software defects to improve system stability, data accuracy, and the reliability of business operations.
- Enhanced the user interface and experience across web, desktop, and mobile applications, improving usability and workflow consistency.
- Collaborated with the development team on feature design, code improvements, testing, and deployment readiness.

Software Engineer Trainee · 4matex Information Technology *Feb 2026 to Apr 2026*

- Completed an intensive Low Code training program with OutSystems, covering application development, database integration, and UI design.
- Built multiple functional business applications in a condensed training cycle, working with Agile workflows in a team.

KEY PROJECTS

PolarisX ERP · **Backend Engineer**

Stack: Sanic, MySQL, SQLAlchemy, REST APIs, JWT Authentication, Flutter, React, Electron

- Worked on the PolarisX backend, an ERP platform for enterprise clients, with React + Electron desktop and Flutter mobile clients.
- Integrated PolarisX with client systems and managed data pipelines between enterprise data sources.
- Website to download PolarisX and see its features: https://polarisx.obsidianlabs.space/

Arjan · Smart Health Clinic Management System · **Graduation Project**

Stack: FastAPI, PostgreSQL, SQLAlchemy, Gemini API, React, TypeScript

- Designed the backend architecture on FastAPI and PostgreSQL: 56 tables across 10 domains, over 160 endpoints, JWT authentication, role based access control, and audit logging.
- Covers the full patient journey: appointments, medical records, lab test lifecycle, referrals, and clinical decision support (drug interaction checks).
- Integrated Gemini for AI analysis of lab reports and medical images, alongside a DICOM viewer for radiology imaging.
- Live demo: https://arjan-demo.netlify.app/#/

RIPPLE · **Incremental Build Analysis Engine**

Stack: C++, Graph Algorithms, Topological Sort, Content Hashing

- Finds exactly which files must rebuild after an edit, and in what order, without launching a compiler, using an include graph, content hashing, and Kahn's algorithm, and reports dependency cycles.
- GitHub: https://github.com/leaderbasil/ripple

TECHNICAL SKILLS

Languages: Python, C++

Backend: FastAPI, Sanic, REST APIs, OpenAPI, JWT Authentication, Async Systems, ERP Integration

Databases: MySQL, PostgreSQL, SQLite, SQLAlchemy

Tools: Git, GitHub, Codeberg, OneDev, Postman, Insomnia, cURL, VS Code, Arch Linux (KDE Plasma)

DevOps: Docker, Kubernetes, AWS

Practices: Team Leadership, SDLC, Software Testing, Code Review, Deployment

Additional Technologies: React, Electron, Flutter, JavaScript, TypeScript, Dart, OutSystems (Low Code)

Soft Skills: Content Strategy, Market & Audience Analysis, Technical Communication, Problem Solving

Interests: AI Agents, Backend Architecture, Web Development, ERP Systems, Mobile Development, Continuous Learning

Looking to learn: Go, Echo, C#, .NET

EDUCATION

Bachelor of Science in Computer Science *2023 to 2027*

Al al-Bayt University

LANGUAGES

Arabic (Native) · English (Professional Working Proficiency)